

Amit Kumar Mohanty

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PROFESSIONAL SUMMARY

Data Scientist with 1+ year of experience turning data into business impact through statistical modeling, machine learning, and predictive analytics. Skilled in Python, R, SQL, statistical inference, feature engineering, and end-to-end model development — from exploratory data analysis and hypothesis testing to deploying production models — with a peer-reviewed publication in deep learning. Outside of work, a competitive tennis player and avid sketch artist, pursuits that sharpen discipline, focus, and creativity.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani) July 2026 – July 2028

M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing)

Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning

K.R. Mangalam University July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana May 2025 – Present

- Awarded **“Rookie of the Year 2025”** for surpassing **all performance KPIs** through advanced modeling, data-pipeline optimization, and cross-functional collaboration.
- Trained and evaluated **Agentic AI models** for complex multi-step workflows, using data-driven experimentation and error analysis to improve **task-completion reliability**.
- Contributed to a **web-automation AI agent** that visually interprets on-screen content and autonomously plans and executes multi-step browser actions from natural-language instructions.
- Trained, evaluated, and quality-controlled the agent with a **global team**, curating datasets and running structured, metric-driven test reviews to improve **action accuracy and reliability**.
- Designed and delivered **85%** of self-generated training datasets to production as the **first member of a global team**, applying rigorous data quality checks and statistical validation.

Artificial Intelligence Intern, Infosys – Remote November 2024 – February 2025

- Analyzed conversational datasets and engineered features that boosted **AI response accuracy by 35%** using LangChain and LangSmith.
- Built and validated LLM-driven features that reduced **system latency by 25%**, quantifying impact with clear performance metrics and dashboards.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing **hallucinations by 20%** through statistical comparison of variants.
- Collaborated with **12 engineers** on end-to-end ML workflows and was selected to present at the **Best Project by Indian Interns** initiative.

PROJECTS

Dynamic Screen Companion June 2025 – Present

- Built a **real-time multi-modal assistant** integrating the **Gemini API** to analyze live screen activity and deliver continuous, context-aware feedback.
- Engineered the end-to-end data and inference pipeline from scratch, reducing manual task time by **40%**.
- Optimized OCR pipelines to sustain **95%+ extraction accuracy** across varied screen layouts.

Diabetic Optiscan January 2025 – June 2025

- Built an AI tool that scans retinal (eye) images to detect early signs of diabetic retinopathy, and co-authored a **peer-reviewed research paper** detailing the methodology and performance results.
- Led integration of a **Vision Transformer (ViT-B/16)** on the APTOS dataset, applying statistical analysis of self-attention to capture spatial patterns in retinal images.
- Designed a custom wavelet technique that boosted **detection precision by 50%** over legacy methods.

Currency Detection App for Visually Impaired July 2024 – December 2024

- Trained an image-classification model on **200,000+ banknote images** achieving **96% accuracy**, validated through rigorous evaluation across edge cases.
- Conducted **user interviews with visually impaired individuals** to ground feature decisions in real-world needs.

SKILLS

Languages & Tools: Python, R, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Jupyter, Git

Statistics & ML: Statistical Modeling, Hypothesis Testing, A/B Testing, Regression, Classification, Clustering, Feature Engineering, Predictive Modeling, Time Series, Model Evaluation & Optimization

Data & Visualization: Data Wrangling, EDA, Power BI, Matplotlib, Seaborn, MySQL, MongoDB, Excel, Statistical Analysis, Experiment Design

AI & Cloud: LLMs, Generative AI, RAG, Deep Learning, NLP, Computer Vision, AWS, GCP (Vertex AI)

CERTIFICATIONS

Ducat India – Data Science Professional Training April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

AI/ML for Geodata Analysis – ISRO

Applied ML and deep learning to satellite and remote-sensing imagery for terrain classification and geospatial pattern analysis.